

From Generic Intelligence to Personalized AI: A Tutorial on Foundations of LLM Personalization

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Abstract

Large language models (LLMs) are moving from cloud services into phones, wearables, vehicles, and smart environments, where useful personalization must respond to people, places, routines, and changing intent. Centralized approaches alone cannot satisfy the simultaneous demands of capability, low latency, resource efficiency, and protection of intimate contextual data. This lecture-style tutorial presents a human-centered view of LLM personalization across cloud, edge, and device. We organize personalization methods into two complementary families: prompting-based approaches that condition generation through instructions, retrieval, and memory, and adaptation-based approaches that update representations, parameters, or preferences. We then connect these methods with on-device learning, federated updates, adaptive model routing, and collaborative inference. Rather than treating a person as a static profile, we frame personalization as an ongoing process in which users can inspect, correct, pause, and revoke adaptation. Situated application examples clarify what should remain on device, what can be processed at the edge, and when cloud intelligence is warranted. We conclude with evaluation methods that jointly consider usefulness, agency, privacy, fairness, latency, energy, and robustness in everyday deployments. The tutorial equips the UbiComp/ISWC community to design personalized LLM systems that are technically effective and accountable to the people who live with them.

CCS Concepts

• **Human-centered computing** → **Ubiquitous and mobile computing**.

Keywords

LLM Personalization, Human-Centered AI, Cloud-Edge-Device Collaboration, Ubiquitous Computing, Responsible AI

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1 Introduction

Ubiquitous computing envisioned computation as part of the fabric of everyday life rather than an object that continually demands attention [14]. Large language models (LLMs) make that vision newly actionable: assistants embedded in phones, earbuds, vehicles, homes, and public environments can interpret natural language and help people across changing situations. Yet an assistant that is merely available everywhere is not necessarily useful everywhere. It must understand whose goals matter, which context is appropriate, how preferences evolve, and when it should remain silent. LLM personalization therefore sits at the intersection of user modeling, context-aware interaction, responsible AI, and systems deployment.

Most personalization research assumes a cloud model that receives profiles, histories, or retrieved memories [1, 12]. In pervasive settings, this assumption is often undesirable. Personal data can



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be intimate or concern bystanders; network connectivity and latency fluctuate; and wearables or mobile devices have tight energy, memory, and compute budgets. At the same time, an entirely local model may lack the knowledge and capacity needed for complex tasks. A cloud–edge–device architecture offers a practical middle ground. The device can maintain recent context, private memory, and lightweight adapters; a nearby edge service can coordinate sensing, caching, and low-latency inference; and the cloud can provide general knowledge and high-capacity reasoning. Federated learning, selective synchronization, collaborative inference, and model routing can then distribute learning and computation without treating every signal as uploadable data [2, 6, 9, 16].

This architecture also changes the meaning of personalization. A person should not be reduced to a persistent profile optimized only for prediction accuracy. Human-centered personalization must make adaptation legible and contestable: users should be able to see why a response changed, correct an inference, separate contexts, pause learning, delete memories, and choose whether a request leaves the device. Designers must also account for shared devices, household negotiation, bystander privacy, accessibility, cultural variation, and the risk that an old preference becomes an unwanted identity. Evaluation must consequently extend beyond response quality to agency, trust, fairness, interruption cost, latency, energy, and longitudinal benefit in real settings.

This tutorial organizes these concerns into six connected lecture modules. Part I introduces situated personalization through an everyday scenario. Part II covers prompting-based personalization, including profile-conditioned instructions, retrieval, memory, and context construction. Part III covers adaptation-based personalization, including user representations, parameter-efficient updates, preference optimization, and continual learning. Part IV maps both method families onto device, edge, and cloud capabilities and studies collaboration policies under resource and privacy constraints. Part V covers human-centered data practices and evaluation, emphasizing consent, feedback, longitudinal change, and field deployment. Part VI synthesizes situated applications and open research questions. The result is a systems-and-interaction perspective that helps UbiComp/ISWC participants design personalized LLMs around human needs rather than model capability alone.

2 Organizers and Instructors

The seven organizers bring complementary expertise in personalized LLMs, user modeling, efficient AI systems, knowledge-enhanced learning, and real-world data mining. They will jointly deliver the lecture modules and situated case analyses in person.

Ruijie Wang is a Professor at Beihang University. His research focuses on trustworthy and efficient AI, large language models, graph learning, temporal data mining, and structured knowledge integration. He has published more than 40 papers at venues including NeurIPS, ACL, KDD, WWW, SIGIR, and UbiComp, and received the IEEE DCOS Best Paper Award. He will connect personalization methods with efficient and responsible deployment.

Qingkai Zeng is an Associate Professor at Nankai University. His research covers text mining, knowledge-enhanced LLMs, and knowledge graph reasoning, with an emphasis on reliable and interpretable knowledge-intensive systems. He has published more than 30 papers at venues such as KDD, WWW, ACL, EMNLP, and

ICLR and has served as a PC member or Area Chair for major conferences.

Xuefei Wang is a Ph.D. student at Beihang University. His research focuses on LLM personalization and LLM system implementation. He contributes implementation experience with personalization pipelines and deployment trade-offs.

Yuhang Wang is an incoming Ph.D. student at Beihang University. Her research focuses on LLM personalization and continual learning. She has published at KDD and ACL and has served on the program committee of NeurIPS. She contributes the perspective of evolving user preferences and continual on-device adaptation.

Li Sun is an Associate Professor at Beijing University of Posts and Telecommunications. His research focuses on large language models, graph representation learning, and knowledge mining. He has published over 70 papers at venues including ICML, NeurIPS, ICLR, KDD, WWW, AAAI, TPAMI, TKDE, TOIS, and TWEB, and received the ACM CIKM 2022 Best Paper Award. He contributes algorithmic perspectives on efficient adaptation and knowledge-enhanced personalization.

Jianxin Li is a Professor at Beihang University. His research interests include social network analysis, trustworthy machine learning, and large-scale data mining. He has published extensively in TKDE, TDSC, TOIS, TKDD, KDD, AAAI, and WWW, and received the AAAI 2021 Best Paper Award and DependSys 2017 Best Paper Award. He contributes expertise in trustworthy learning and large-scale systems.

Phillip S. Yu is a Distinguished Professor and the Wexler Chair in Information Technology at the University of Illinois Chicago. He is a Fellow of ACM, IEEE, and AAAS. His research spans data mining, graph learning, user behavior modeling, and multimodal LLMs. He received the ACM SIGKDD Innovation Award, the IEEE Computer Society Technical Achievement Award, and multiple Test-of-Time awards, and contributes foundational expertise in user modeling and large-scale analytics.

3 Tutorial Details

3.1 Three-Hour Schedule

The tutorial is delivered entirely in a lecture style, progressing from conceptual foundations to methods, systems, evaluation, and applications. The planned schedule is:

- **Part I: People, Context, and Personalization (20 min).** From generic assistants to situated, evolving, and user-governed experiences; problem definitions and a running urban-mobility scenario.
- **Part II: Prompting-Based Personalization (30 min).** Profile-conditioned instructions, retrieval and memory, demonstration selection, prompt optimization, and context management.
- **Part III: Adaptation-Based Personalization (30 min).** User representations, parameter-efficient adapters, supervised fine-tuning, preference optimization, federated adaptation, and continual learning.
- **Part IV: Cloud–Edge–Device Collaboration (40 min).** Placement of data, memory, models, and computation; split and cascaded inference; routing under latency, energy, connectivity, privacy, and quality constraints.
- **Break (10 min).**

- **Part V: Human-Centered Data and Evaluation (30 min).** Consent and provenance, shared-device and bystander scenarios, inspectable memories, user correction and revocation, and evaluation in longitudinal field use.
- **Part VI: Situated Applications and Open Challenges (15 min).** Mobile well-being, shared-home, and urban-mobility examples; unresolved questions spanning people, methods, and systems.
- **Synthesis (5 min).** Main takeaways and a research agenda for the UbiComp/ISWC community.

3.2 Part II: Prompting-Based Personalization

Prompting-based methods personalize generation without changing the base model parameters. Profile instructions describe stable preferences or communication needs. Retrieval selects relevant episodes, facts, or summaries from a user’s history [11], while demonstration selection exposes the model to representative prior interactions. We cover memory construction, prompt templates, personalized exemplars, context compression, and prompt optimization [4, 13]. The lecture compares their low update cost and inspectability with limitations in context length, retrieval error, stale memories, prompt injection, and inadvertent disclosure. It also distinguishes what belongs in a durable profile, an episodic memory, or transient situational context.

3.3 Part III: Adaptation-Based Personalization

Adaptation-based methods modify representations, model parameters, or behavioral objectives. We introduce learned user embeddings and soft prompts, supervised fine-tuning, parameter-efficient adapters, and personalized heads [5, 7]. We then examine preference optimization from explicit or implicit feedback [8, 10], continual learning for changing routines, and federated or selectively shared updates for distributed users. The lecture emphasizes data and computation requirements, update stability, catastrophic forgetting, rollback and deletion, cross-user interference, and the difficulty of making parameter-level adaptation legible. Prompting-based and adaptation-based approaches are compared as complementary choices rather than mutually exclusive alternatives.

3.4 Part IV: Collaboration Across Cloud, Edge, and Device

This part introduces a three-tier design space. The *device* is closest to the person and can retain sensitive raw context, recent interaction state, and small adapters. The *edge* can aggregate nearby signals, cache models, and provide low-latency coordination across a home, vehicle, campus, or venue. The *cloud* supplies high-capacity reasoning, global knowledge, and expensive updates. We compare local-only, cloud-only, cascaded, split, and opportunistic execution; federated or selectively shared updates [6]; and collaborative personalization in which a cloud model supports a smaller on-device model [9, 16]. Participants learn to formulate routing as a multi-objective decision over quality, latency, energy, communication, privacy exposure, and user preference. We also discuss graceful degradation during disconnection, deletion propagation, cross-device consistency, and attacks on personalized memories.

3.5 Part V: Human-Centered Data and Evaluation

Personalization data are sparse, socially situated, and time-varying. We distinguish user-, persona-, household-, and population-level signals and examine cold start, context collapse, preference drift, and unequal coverage [3, 15]. A practical lifecycle connects notice and consent to collection, inference, storage, adaptation, explanation, correction, expiration, and deletion. Evaluation similarly spans multiple layers: model quality and calibration; system latency, energy, and robustness; interaction measures such as workload, interruption, trust, and perceived control; and longitudinal outcomes such as sustained utility and changing behavior. Participants will learn why an offline personalized benchmark cannot by itself validate an everyday system, and how mixed methods, experience sampling, diary studies, and field trials can expose harms or benefits that aggregate accuracy hides.

3.6 Lecture Format, Outcomes, and Materials

The tutorial uses a structured lecture format with a consistent progression from concepts and method taxonomies to system architectures, human-centered evaluation, and situated applications. By the end, participants will be able to (1) distinguish prompting-based from adaptation-based personalization; (2) allocate data and computation across device, edge, and cloud; (3) identify requirements for inspectable, correctable, and revocable adaptation; and (4) evaluate both technical performance and human experience. Slides, lecture notes, architecture comparison tables, and an annotated reading list will be openly available from the tutorial website. Participants need no programming experience, account, special hardware, or pre-tutorial preparation.

4 Situated Application Cases

Three lecture cases keep technical mechanisms grounded in human experience. First, a *mobile well-being assistant* combines a watch, phone, and conversational model. Raw physiological signals and short-term routines remain on the device; an edge node supports low-latency activity recognition; and cloud reasoning is requested only for complex reflection or general health information. The case illustrates clinically inappropriate inference, notification timing, deletion, and whether a user’s decision not to engage should itself become a persistent signal. We contrast an accuracy-maximizing design with one that privileges calm interaction, data minimization, and an explicit non-personalized mode.

Second, a *shared-home assistant* demonstrates that “the user” is often plural. A model may encounter household members, guests, children, care workers, and bystanders through the same microphones and displays. A single global memory can collapse incompatible roles and disclose one person’s preference to another. We examine scoped memories, presence-aware inference, local identity boundaries, visible sensing cues, and household negotiation. An edge hub can retain room-level context without sending raw audio to the cloud, but it also becomes a consequential point of control. The lecture examines who can inspect or delete a shared memory and how the system should behave when consent cannot be established.

Third, an *urban mobility assistant* follows a person across earbuds, phone, vehicle, transit infrastructure, and intermittent connectivity. The system must decide whether a request can be handled locally, whether a nearby edge service is trustworthy, and whether cloud reasoning adds enough value to justify delay and disclosure. Preferences that are benign in one context can become sensitive in another: a quiet-route preference may reveal health status, safety concerns, or religious practice. We analyze alternative routing policies under network loss, device handover, a sudden preference change, and a bystander interaction. Together, the cases illustrate that placement and interaction design are inseparable: where personalization happens shapes what a system can know, how quickly it can respond, and how much control a person can meaningfully exercise.

5 Why UbiComp/ISWC

The tutorial addresses a timely convergence in the UbiComp/ISWC community. Foundation models increasingly mediate sensing and interaction, while capable small models and hardware accelerators make local adaptation feasible. Researchers now need a shared language spanning model personalization, mobile and edge systems, HCI, privacy, and field methods. Existing LLM tutorials commonly emphasize algorithms or recommendation; edge tutorials commonly emphasize efficiency. Our contribution is to connect these perspectives around a human-centered question: *how should an intelligent environment adapt to a person without taking control away from them?* The tutorial will seed collaborations across technical and empirical communities, provide a reusable taxonomy and evaluation framework, and surface open problems that require both systems innovation and long-term study with people.

6 Related Tutorials

- **Tutorial on Large Language Models for Recommendation**
 - **Presenters:** Wenyue Hua, Lei Li, Shuyuan Xu, Li Chen, and Yongfeng Zhang
 - **Conference:** RecSys '23, September 18–22, 2023, Singapore
 - **Connection:** This tutorial introduces large language models for recommendation, covering datasets, models, evaluation, toolkits, and real-world systems.
 - **Differences:** That tutorial focuses on recommendation as a downstream application. Ours spans assistants, wearables, health, mobility, and smart environments, with explicit attention to human agency and cloud–edge–device deployment.
- **Large Language Models for Recommendation: Progresses and Future Directions**
 - **Presenters:** Jizhi Zhang, Keqin Bao, Yang Zhang, Wenjie Wang, Fuli Feng, and Xiangnan He
 - **Conference:** WWW '24, May 13–17, 2024, Singapore
 - **Connection:** This tutorial presents recent advances in using large language models for recommendation, which is closely related to personalized generation and user-aware modeling.
 - **Differences:** That tutorial centers on recommendation algorithms. Ours connects personalization methods to situated interaction, collaborative inference, resource constraints, privacy, and longitudinal field evaluation.

7 Tutorial Participation

Target Audience and Expected Reach. The intended audience includes students, researchers, designers, and practitioners in ubiquitous computing, HCI, mobile and edge systems, wearables, user modeling, and LLMs. Basic familiarity with machine learning or interactive systems is helpful but not required. Because personalized assistants increasingly span phones, wearables, vehicles, homes, health, and education, the topic is relevant to academic and industrial audiences. We will promote the tutorial through the UbiComp/ISWC, HCI, mobile-systems, and responsible-AI communities. We estimate 50–80 participants; participation is not otherwise limited.

Engagement Plan and Expected Impact. The tutorial will be delivered as a coherent sequence of lectures. Conceptual taxonomies, comparative method analysis, system diagrams, and three situated application cases will connect the technical material to usability, agency, privacy, and deployment constraints. Afterward, slides, lecture notes, architecture comparison tables, and an annotated reading list will remain openly accessible through the website.

8 Social Impact Statement

This tutorial is closely aligned with UbiComp/ISWC because personalized LLMs are becoming part of everyday devices and environments. They may improve accessibility, continuity of care, learning, and assistance by adapting to diverse needs and contexts. The same capabilities can intensify surveillance, infer sensitive attributes, expose bystanders, reinforce stereotypes, or make unwanted identities persistent. Cloud–edge–device designs do not automatically solve these problems: local processing can reduce disclosure yet still enable opaque inference, while federated updates can leak information or encode unequal participation. We therefore treat consent, purpose limitation, inspectability, correction, deletion, fairness, accessibility, and graceful non-personalized fallback as first-class design requirements. The tutorial also emphasizes that the person, not engagement or prediction accuracy, should define when personalization is beneficial.

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